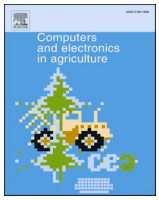
Computers and Electronics in Agriculture 226 (2024) 109426



Contents lists available at ScienceDirect

Computers and Electronics in Agriculture

journal homepage: www.elsevier.com/locate/compag





New method for modeling digital twin behavior perception of cows: Cow daily behavior recognition based on multimodal data

Yi Zhang [a,b], Yu Zhang [a,*], Hailong Jiang [a], Haitao Du [c], Aidi Xue [a,d], Weizheng Shen [a,*]

[a] College of Electrical and Information, Northeast Agricultural University, Harbin 150030, China
[b] College of Information Engineering, East University of Heilongjiang, Harbin 150066, China
[c] Dairy Association of Heilongjiang Province, Harbin 150020, China
[d] Harbin Electric Machinery Company Limited, Harbin 150046, China

ARTICLE INFO

Keywords:
Multimodal classification
Digital twin
Dairy cow
Behavior recognition
Transformer

ABSTRACT

The cow digital shadow reflects the behavior, health condition, and productivity of cows, playing a crucial role in ensuring animal welfare, increasing individual productivity, and improving breeding efficiency. To fully utilize the existing multimodal data on farms and build a cow digital shadow with rich behavioral information, this study proposes a multimodal data fusion algorithm for recognizing cow behaviors such as drinking, feeding, lying, standing, and walking. This algorithm leverages the strengths of different data modalities, complementing each other, and enhances the performance of the cow behavior classification model. The algorithm integrates motion sensor and video data, collected by custom-made collars with inertial measurement units (IMUs) sensors placed at the top of the cow's neck and cameras in the barn, using EfficientNet V2 S, BiLSTM, and Transformer networks. Experimental results demonstrate recognition accuracies of 98.80 %, precision of 97.15 %, and recall rates of 96.93 %, showing significant improvements over single-modal data behavior recognition algorithms. This method maximizes the utility of existing multimodal data to generate a cow digital shadow with detailed behavioral information, enhancing the modeling and simulation element of the cow digital twin architecture and laying the foundation for developing a comprehensive cow behavior data model.

1. Introduction

The construction of a cow digital shadow (Zhang et al., 2023) enables a comprehensive understanding of cow behavior, health status, and productivity, playing a crucial role in ensuring animal welfare (Tucker et al., 2021), enhancing individual production, and improving breeding efficiency (Halachmi et al., 2019). The cow digital shadow encompasses various aspects of cow information, with the collection and analysis of cow behavior data being significant for understanding health status, estrus conditions, and disease occurrences (Martin et al., 2021). Advancements in sensor technology, artificial intelligence, and big data have enabled real-time identification of cow behavior (García et al., 2020). Currently, farms employ a variety of sensing devices, such as accelerometers, cameras, cow sensors, and scales, to monitor metrics including cow activity, diet, and physical condition, as well as to detect diseases and estrus (Stygar et al., 2021). These behavior recognition methods can be classified into wearable and non-contact types. Wearable methods use sensors like accelerometers, inertial measurement units, pedometers, and magnetometers to collect data on various behaviors and employ machine learning techniques for behavior identification and tracking. Non-contact methods leverage cameras and machine vision techniques to recognize animal behaviors without disturbing them, identifying basic motor behaviors of cows (Liu et al., 2023b). Currently, data used in these two methods are heterogeneous in form and independent in approach. Therefore, fully utilizing data generated by existing devices, considering the characteristics and advantages of different modal data, and improving the accuracy and precision of cow behavior recognition methods to construct more information-rich cow digital shadows is a key issue this research aims to address.

The sensing method for cow behavior using wearable devices, such as collars, employs various sensor data to automatically identify cow behaviors. This reduces manual labor, improves management efficiency, and minimizes human interference and errors (Chen et al., 2022). Early researchers used pedometers as an auxiliary means to detect cow estrus behavior (Peter and Bosu, 1986). With the advent of computer

* Corresponding authors.
E-mail addresses: zhangyu1900@neau.edu.cn (Y. Zhang), wzshen@neau.edu.cn (W. Shen).

https://doi.org/10.1016/j.compag.2024.109426
Received 13 June 2024; Received in revised form 26 August 2024; Accepted 30 August 2024
Available online 5 September 2024
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technology, accelerometers and personal computers were employed to record and measure cow activity levels (Müller and Schrader, 2003). In recent years, researchers have increasingly adopted machine learning and deep learning methods to analyze cow behavior. Shen et al. (Shen et al., 2020) identified cow feeding and rumination behaviors using triaxial accelerometer data combined with machine learning algorithms. Tian et al. (Tian et al., 2021) collected cow behavior data using multiple sensors and applied KNN and RF models to identify seven behaviors, achieving an average recognition accuracy of 98.51 % with the KNN-RF fusion model. Liu et al. used a Fully Convolutional Network (FCN) algorithm to automatically classify seven cow behavior patterns through Inertial Measurement Units (IMU), attaining over 80 % recognition accuracy (Liu et al., 2023a). However, cow behavior recognition based on wearable devices faces issues, such as insufficient data utilization, leading to poor distinction of certain behaviors when solely relying on sensor data. Wang et al.'s research indicated that feeding, standing, and moving are the most difficult behaviors to distinguish in classic machine learning models (Wang et al., 2018). Moreover, the position of sensor placement significantly affects classification results. Benaissa et al. (Benaissa et al., 2019) used combined data from accelerometers mounted on the neck and legs to classify cow behaviors, achieving high accuracy (80–99 %) and high sensitivity (87–99 %) in classifying lying, standing, and feeding behaviors. However, placing multiple sensors on different parts of the cow to achieve high-precision behavior distinction increases the risk of stress reactions. In summary, cow behavior perception methods based on wearable devices such as collars exhibit high data accuracy, strong real-time performance, and excellent device stability. However, reliance on a single device may lead to insufficient differentiation of certain behaviors, while the use of multiple devices increases stress on the cows, negatively impacting cost control and animal welfare.

Non-contact behavior sensing methods based on cameras use RGB or grayscale cow images or video data and employ deep learning algorithms to classify cow behaviors. Deep learning offers the advantages of being non-contact and highly precise, accurately identifying animal behavior using only image and video data (Wang et al., 2022). Yin et al. (Yin et al., 2020) utilized EfficientNet for spatial feature extraction, BiFPN for feature fusion, and the BiLSTM attention mechanism for temporal aggregation, achieving a recognition accuracy of 97.87 % and an average frame rate of 134 frames per second. However, standing and walking behaviors are difficult to distinguish in natural environments since cow behavior is often a complex mixed state, with recognition easily affected by posture and environment. These two behaviors are visually very similar, making recognition challenging. Fuentes et al. (Fuentes et al., 2020) proposed a deep learning-based framework for automatically recognizing cow behaviors in real farm environments, including 15 different levels of behaviors such as feeding, walking, resting, standing, and lying down. This framework combines spatial and temporal information to monitor cow behaviors in real-time and provide behavior localization. However, algorithm performance may be affected by long distances and occlusions. In summary, camera-based non-contact behavior perception methods of dairy cows offer the advantages of low stress on animals, wide coverage, and high accuracy. However, these methods are susceptible to environmental factors such as distance, lighting, and occlusion, and it is challenging to differentiate visually similar behaviors.

This paper proposes a multimodal fusion method for cow behavior recognition based on sensor and video data. By integrating multimodal data, the advantages of different data types are harnessed, overcoming the shortcomings of single-modal data in cow behavior recognition. Combining sensor data improves the recognition accuracy of visually similar behaviors, such as cow walking and standing, while integrating video data addresses the recognition issue for behaviors like standing and lying down, which appear similar in sensor data. This method effectively utilizes the multimodal data generated by existing farm equipment to create behavior-rich cow digital shadows, laying the foundation for building a comprehensive model of cow behavior data. Additionally, this method minimizes cow stress by using only a custom-made collar at the top of the cow's neck. The collar, which contains cow identity information, resolves the identity recognition and target-tracking issues encountered in non-contact methods under conditions of distant views and small targets.

2. Materials and methods

2.1. Data collection

The sensor and video data used in this study were collected at the Acheng experimental base of Northeast Agricultural University from May 27 to June 13, 2022. The subjects were five healthy, non-lactating Holstein cows, aged 3 ± 0.5 years and weighing 410 ± 40 kg. The experimental procedures adhered to the relevant provisions of the "Regulations on the Administration of Laboratory Animals of the People's Republic of China" (Zhang et al., 2023). A top view schematic of the multimodal data acquisition system is shown in Fig. 1.

This system can simultaneously collect collar sensor data and video data for the five experimental cows. By continuously recording the behavior data of the experimental subjects, video and sensor data were collected under different time periods and lighting condition.

2.1.1. Sensor data collection

A custom-made collar integrated with IMUs was employed to collect real-time motion sensor data on the drinking, feeding, lying, standing, and walking behaviors of the experimental subjects. The collars, in conjunction with anchor points, constituted a sensor signal acquisition system utilizing ultra-wideband communication. The sensor signal acquisition system used in this study comprises six UWB signal receiving anchor points connected in series via the RS422 bus. One anchor point serves as the master anchor, connected to the serial port device server via RS422. The serial port server communicates with the ranch server using the TCP protocol through a twisted pair. The collar worn on each cow's neck collects sensor data at a frequency of 4 Hz, including acceleration and angular velocity data along three axes. This data is transmitted to the master anchor every 90 s via UWB signal. The master anchor, with the help of the serial port device server, sends these signals to the ranch server, where they are ultimately stored in CSV format. Each row of data includes the packet arrival time, packet number, and the acceleration and angular velocity data in three directions.

2.1.2. Video data collection

To meticulously record the video data of the drinking, feeding, lying, standing, and walking behaviors of the experimental subjects, a video acquisition system was specifically designed for this study. Five network cameras (CS-C1C, Hikvision, Hangzhou, China) were installed on the steel beams of the cowshed, 3 m in front and 2.6 m above the feeding area, capturing footage at a downward angle. Each camera was positioned directly in front of an individual cow to record its behavior, with overlapping fields of view to provide multiple perspectives. Each camera was equipped with a 128 GB Micro SD card. Researchers copied the videos weekly to the ranch server and cleared the SD cards. The video resolution was 1920 × 1080, allowing clear observation of various cow behaviors.

2.1.3. Dataset construction

In this experiment, three experienced keepers observed and recorded the behaviors of drinking, feeding, lying, standing, and walking of five cows. Subsequently, 228 video segments were manually selected, with each behavior lasting 5000 s, equating to 75,000 frames per behavior and 375,000 frames in total. Specifically, there were 35 segments of drinking, 15 segments of feeding, 26 segments of lying, 46 segments of standing, and 105 segments of walking, averaging 109.6 s per segment. The detailed specifications of the video data employed in this

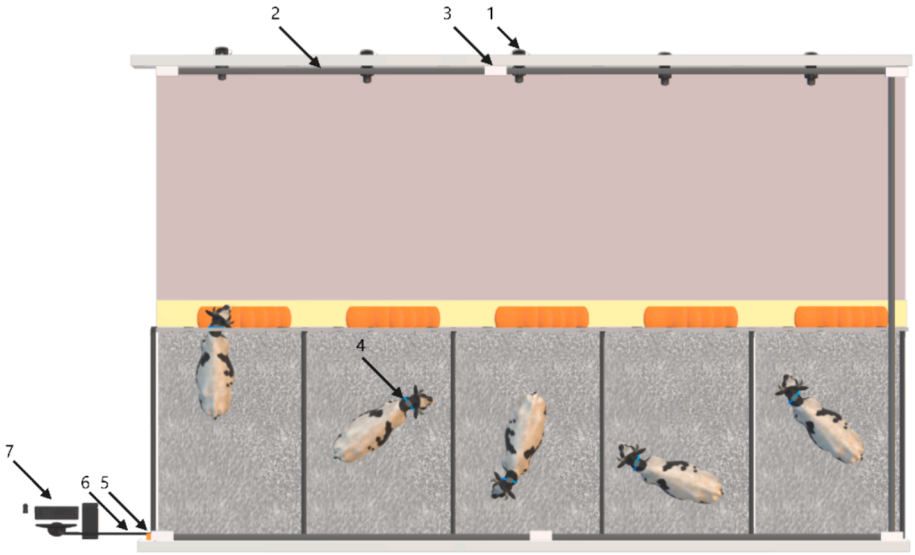


Fig. 1. Top view schematic diagram of the measurement system. Camera (1), RS422 Bus (2), Anchor Point (3), Collar (4), Serial Port Device Server (5), Twisted Pair (6), Ranch Server (7).

experiment are provided in Table 1. Due to pen size constraints, the walking duration was generally short, while the standing still duration was also brief due to the cows' natural behaviors. Additionally, lying behaviors were observed for extended periods, predominantly during nighttime. Drinking behaviors occurred for shorter durations, while feeding behaviors were more concentrated and occurred at similar times. For consistent behavior representation and to avoid model bias from uneven video lengths, 5000 s of video were extracted for each action. Sensor data segments matching the video duration were then extracted based on the video's start time and its length.

Given the sensor's sampling frequency of 4 Hz, video decomposition technology was employed to uniformly extract 4 frames per second, resulting in 20,000 frames for each behavior and a total of 100,000 video snapshots. At the same time, the frame resolution was adjusted to 640 × 360 pixels. Using object detection technology, the cows' positions in the video frames were detected and cropped. Examples of dataset images are shown in Fig. 2. The selected videos adhered to the following standards:

1. The dataset should encompass diverse lighting conditions, camera angles, and cow behaviors to ensure sample diversity and improve the model's generalization ability.
2. All selected behavior segments were ongoing cow behaviors, excluding the initiation and termination portions.
3. Drinking behavior was defined from when the cow's tongue first contacted the water surface to when the cow's head left the drinking area.
4. Walking behavior was defined from the first leg movement to the final hooffall.
5. All behavior segments were truncated from one second before the behavior start to one second after the behavior ended.

Table 1
Video Data of Cow Behaviors Used in This Study.

Behaviors	Video segment number/segment	Average Video duration/s	Frame rate/fps	Total frames/ frames
Drinking	36	138.9	15	75,000
Feeding	15	333.3	15	75,000
Lying	26	192.3	15	75,000
Standing	46	108.7	15	75,000
Walking	105	47.6	15	75,000

The dataset was split into training and test sets in a 9:1 ratio, with the training set containing 90,000 cropped cow images and the test set containing 10,000 images. The folders for the two sets are named "train" and "val" respectively, and their internal structures are consistent. Each behavior is assigned a number, and a corresponding numbered folder is created. These numbered folders contain all the data for the same behavior. Subdirectories are named after each video segment's filename, and these subdirectories contain all the cropped images of cows from that video segment, along with the corresponding raw CSV files from the cow sensor.

As illustrated in Fig. 2, each row contains keyframe croping images depicting the same behavior from video segments in the dataset. Behavior numbered 0 represents drinking, behavior numbered 1 represents feeding, behavior numbered 2 represents lying, behavior numbered 3 represents standing, and behavior numbered 4 represents walking. The columns labeled a, b, and c correspond to different keyframe croping images. Simultaneously, the volume and distribution of sensor data within the dataset are aligned with the keyframe cropped images, based on the timing and duration of the observed behaviors.

2.2. Overall Technical Route

To effectively integrate video and sensor data modalities for improved classification results, the processing procedure is depicted in Fig. 3. This process is broadly divided into two stages: feature extraction and feature fusion. The feature extraction for sensor data is relatively straightforward: the model is pre-trained using BiLSTM to identify the most effective recognition model. This pre-trained model is subsequently used to extract feature vectors from the sensor data.

In contrast, the feature extraction for video data involves several steps:, the feature extraction for video data involves several steps: 1. Using the OpenCV library, frames numbered 0, 4, 8, and 12 per second are extracted from a video with a frame rate of 15 FPS to serve as dataset sample frames. To reduce convolutional computation, the dimensions of these sample frames are scaled down to one-third of their original size, resulting in a resolution of 640 × 360 pixels. 2. Utilizing YoloV7 Tiny, a target detection model is trained with 3000 images labeled with five cow identifiers. This detection model is then applied to the sample frames obtained in step 1, cropping the detected cow images and placing them into folders named according to cow identifiers c1 to c5. 3. The EfficientNet V2 S network is employed to classify the cropped cow images from step 2 into five behavioral categories. The resulting pre-trained

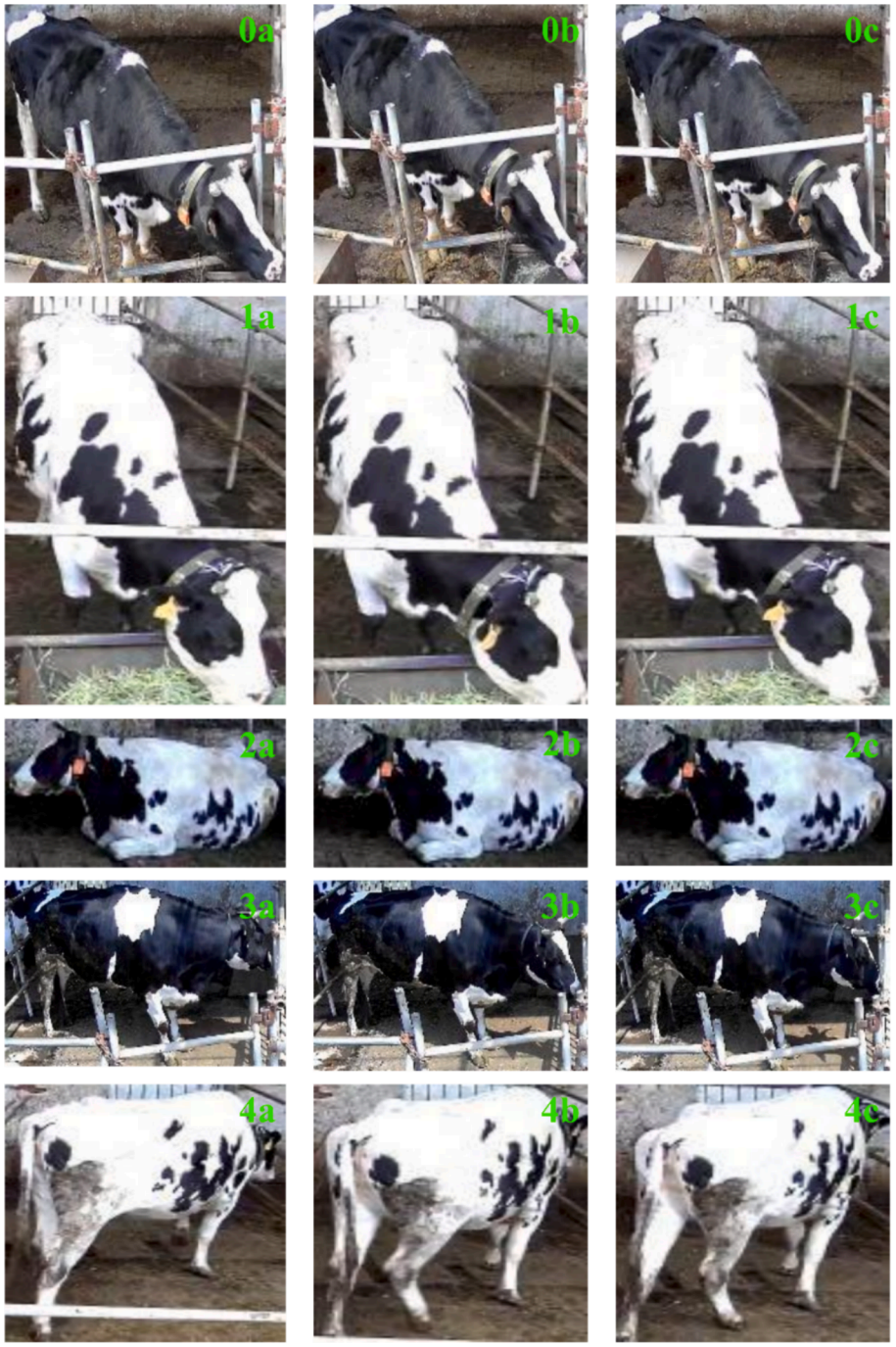


Fig. 2. Dataset sample examples.

model is used to extract behavioral feature tensors from the cow images.

Once the image feature tensors from the video data and the feature vectors from the sensor data are prepared, they are fused and fed into a Transformer Encoder. The output is then classified to produce the final classification results.

2.3. Preprocessing

2.3.1. Key frame extraction

To effectively extract motion information of cows from video footage while balancing training costs and achieving modal alignment with sensor data, it is essential to extract key frames. This study employs the OpenCV library to read videos and save specified frames by traversing each frame. The program captures one frame and skips the next four frames three times, then skips three frames after the fourth capture. This method ensures the uniform saving of four images per second. Experiments demonstrate that, under stable frame rate conditions, this approach successfully saves four frames per second. Using this method, a total of 100,000 key frames representing five behaviors of the experimental subjects were extracted, with 20,000 frames for each behavior.

2.3.2. Cow image cropping

To extract regions of interest from video frames and accurately identify cows, a Yolo V7 network based on PyTorch is trained. This ensures that cows occupy more than 80 % of the image. The trained model is used to capture snapshots of the video frames. A dataset was created according to the PASCAL VOC 2007 standard, comprising 3,000 images and 7,514 targets across five categories, with a training-to-test

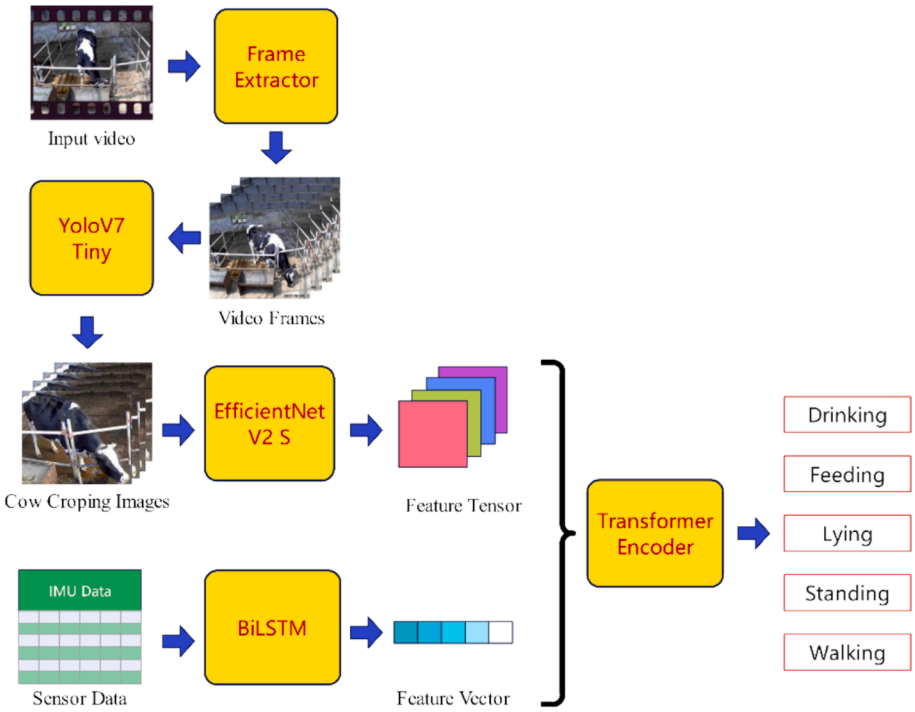


Fig. 3. Multimodal Behavior Recognition Flowchart.

set ratio of 9:1. Prior to training, general pre-trained weights are loaded to avoid overly random backbone network parameters. The model underwent 300 training epochs, with the best performance observed at the end of the 290th epoch. At this stage, the loss rate was 0.023 for the training set and 0.022 for the validation set. When evaluating targets in the validation set, the mean Average Precision (mAP) reached 98.60 % at an Intersection over Union (IoU) of 0.5. This model detects and crops images of cows in the key frames, ultimately obtaining the image data for cow behavior.

2.4. Pretraining

This study considers the characteristics of different data types by pretraining separate models for motion image and sensor data. These pretrained models are then used to extract features from their respective data types, to facilitate feature fusion. The pre-training processes for these two types of data are described below.

2.4.1. Feature extraction for cow images

EfficientNet V2 (Tan and Le, 2021), an improved model based on EfficientNet V1, introduces progressive learning strategies and adaptive regularization, which expedite training and enhance both inference and training speeds. To unify the dimensions of motion image data and sensor data and to enhance the training efficiency of the fusion model, this study utilizes EfficientNet V2 for the pretraining and feature extraction of cow motion image data. Given the task's complexity and the images' characteristics, EfficientNet V2 S was selected as the backbone network and pretrained on the dataset. Each of the five behaviors consists of 20,000 images, partitioned into training and validation sets in a 9:1 ratio.

During the five-class pretraining using the EfficientNet V2 S backbone network, cow keyframe screenshots were resized to 256 × 256 pixels, using a batch size of 256 and an input size of 100. To enhance model generalization, images were horizontally flipped with a 50 % probability. Pretraining lasted for 300 epochs, with the highest validation accuracy achieved at the 117th epoch, yielding an overall recognition accuracy of 89.46 % for the five behaviors. This model was then used to extract image features from the dataset, resulting in image feature tensors with dimensions (256, 8, 8) before the fully connected layer.

2.4.2. Feature extraction for sensor data

Bidirectional Recurrent Neural Networks (BRNN) can leverage both past and future information for predictions, without requiring pre-set delay parameters. By training the network in both temporal directions simultaneously, BRNNs can better utilize available input information, achieving superior results in practical applications (Schuster and Paliwal, 1997). BiLSTM, an extension of this concept, applies LSTM units within bidirectional RNNs, enhancing the model's ability to handle long-term dependencies and contextual information. The collected sensor data in this study include acceleration and angular velocity in three directions, representing changes in three-dimensional space for the experimental subjects' different actions, and thus possess time-series characteristics. Therefore, BiLSTM is highly suitable for classification. The study extracted 100,000 corresponding sensor data points at the key frame time points, also divided into training and validation sets in a 9:1 ratio. During the data preprocessing stage, all data were standardized using StandardScaler.

After multiple trial training sessions, the optimal BiLSTM configuration was selected: 7 hidden layers with 128 nodes per layer, a batch size of 32, and a sequence length of 16. The input data included acceleration and angular velocity in three directions. To enhance the model's generalization ability and avoid overfitting, a dropout rate of 0.5 and an L2 regularization coefficient (weight decay) of 0.0002 were used, with Adam as the optimizer. Data at each time step were classified over 200 training epochs, with the highest validation accuracy observed at the 49th epoch, achieving an average recognition accuracy of 96.83 % for the five behaviors. This model was used to extract sensor data features from the dataset, resulting in sensor feature vectors with a length of 256 before the fully connected layer.

2.5. Multimodal data fusion

2.5.1. Modality alignment

Modality alignment refers to the process of ensuring that information from different data sources is precisely synchronized in both temporal and spatial dimensions during multimodal data fusion. This alignment is critical for optimizing system design, enhancing the accuracy and robustness of recognition algorithms, and improving the performance of model training and inference. In this study, video data is captured at 15 frames per second, while sensor data is recorded at 4 data points per second. Aligning these two modalities is a crucial issue addressed in this research. Temporally, when extracting key frames from the video, only frames numbered 0, 4, 8, and 12 from each second are retained as key frames, adjusting the video frame sampling frequency to align with the sensor's 4 Hz frequency.

For data feature alignment, the feature tensor for the keyframe screenshot data is shaped (256, 8, 8). To preserve global features in the tensor, global average pooling is applied to obtain a feature vector of length 256. To retain significant features, max pooling is used to obtain another feature vector of the same length. These vectors are concatenated and standardized, and then converted into a feature vector of length 128 using a fully connected layer. For the sensor feature vector, initially of length 256, it is first standardized and then transformed into a 128-length vector using a fully connected layer, completing the feature alignment between video and sensor data.

2.5.2. Modality fusion

The input to a Transformer can consist of one or more token sequences and their associated attributes, enabling it to perform multimodal learning tasks without modifying its architecture (Devlin et al., 2018). Consequently, this study employs a Transformer to achieve deep modality fusion, as illustrated in Fig. 4.

Prior to multimodal data feature fusion, feature extraction is conducted on single-modality sequence data. Firstly, a pre-trained Efficient V2 S model extracts features from each frame in the trimmed keyframe sequence, producing a feature tensor before the fully connected layer. This tensor subsequently undergoes global average pooling and maximum pooling. The pooling results are concatenated and stacked to form the feature tensor of the keyframe sequence, which is then normalized. This normalized tensor is fed into the fully connected layer to obtain the keyframe image features. Concurrently, a pre-trained BiLSTM model extracts features from the sensor data. The features are first normalized and then fed into the fully connected layer to reshape them, resulting in the sensor data features. Finally, the image features and sensor features, now sharing the same shape, are concatenated and used as input to the Transformer Encoder, thereby achieving deep feature fusion across different modalities.

2.5.3. Multimodal training

To achieve daily behavior recognition of cows based on multimodal data, a training network was constructed using the PyTorch library. The dataset was divided into training and testing sets, and the processing and transformation of all data sets during model training can be divided into five steps:

1 Processing each cropped keyframe image to obtain features:

$$i_f = FC1(i_n([GAP(f(i)) \oplus MP(f(i))])) \tag{1}$$

Where i represents the cropped keyframe, $f(i)$ denotes the process of extracting image features using the EfficientNet V2 pretrained model, $GAP(x)$ represents global average pooling, $MP(x)$ represents max pooling, $\oplus$ denotes the vector concatenation operation, $i_n(x)$ represents the image normalization layer, and $FC1(x)$ represents the first fully connected layer.

2 Processing sensor data to obtain features:

$$s_f = FC2(s_n(fs(s))) \tag{2}$$

Where s represents the sensor data, $fs(x)$ denotes the process of extracting sensor data features using the BiLSTM pretrained model, $s_n(x)$ represents the normalization layer, and $FC2(x)$ represents the second fully connected layer.

3 Concatenating the features from both modalities:

$$c_f = cat(i_f, s_f) \tag{3}$$

Where i_f is the image feature, s_f is the sensor feature, and $cat(x, y)$ denotes the concatenation operation along the last dimension.

4 Inputting the concatenated features into the Transformer encoder:

$$t = t_e(c_f) \tag{4}$$

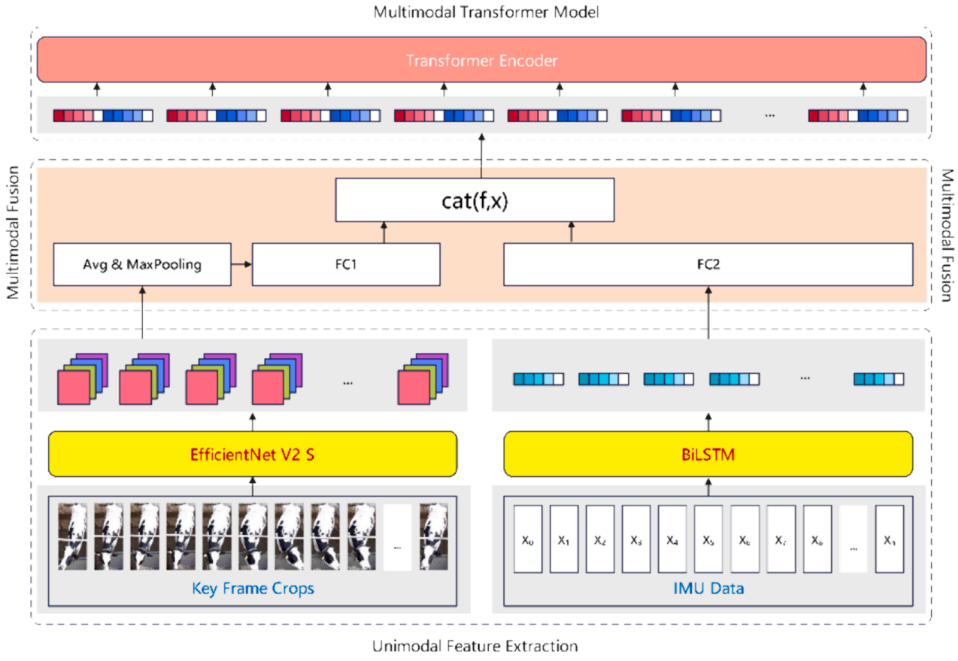


Fig. 4. Multimodal Data Fusion Method.

Where c_f represents the concatenated features, and $t_e(x)$ represents the Transformer encoder.

5 Passing the output of the encoder at the final time step through the final fully connected layer to obtain the result:

$$out = FC3(s(t)) \tag{5}$$

Where t represents the output of the Transformer encoder, $s(x)$ represents the operation of obtaining the tensor at the final time step, and $FC3(x)$ represents the third fully connected layer.

This study utilizes a 2-layer Transformer encoder to classify cow behavior following multimodal data fusion. The model expects 256 input features, with the first 128 representing image features and the remaining 128 representing sensor features. It employs 8 multi-head attention mechanisms, a learning rate of 0.001, the Adam optimizer, and a batch size of 32, and is trained over 20 epochs. The highest accuracy, with an average of 98.8 % across five behaviors, is achieved in the first epoch.

2.6. Evaluation Indicators

To evaluate the algorithm's performance in recognizing different behaviors, this study employs the confusion matrix, Precision, Recall, and Accuracy as performance metrics. The confusion matrix provides a clearer understanding of classification errors. Precision and Recall are scaled between 0 and 1. Precision indicates the proportion of true positive predictions among all positive predictions, while Recall denotes the proportion of true positive samples among all actual positive samples in the test set. Accuracy measures the proportion of correctly classified samples out of the total number of samples. The calculation methods for these metrics are detailed in equations (6) through (8).

$$accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{6}$$

$$precision = \frac{TP}{TP + FP} \tag{7}$$

$$recall = \frac{TP}{TP + FN} \tag{8}$$

Where TP (True Positive) indicates predictions that are positive and correct; FN (False Negative) indicates predictions that are negative but incorrect; TN (True Negative) indicates predictions that are negative and correct; FP (False Positive) indicates predictions that are positive but incorrect.

3. Results and discussion

3.1. Performance of different feature fusion methods

In this study, three different feature fusion methods were employed during the feature fusion stage: mean fusion, max fusion, and feature concatenation fusion.

Mean fusion calculates the average of the features from different modalities, as represented by the following formula:

$$combied_features = \frac{features_1 + features_2}{2} \tag{9}$$

Max fusion takes the maximum value of the features from different modalities, as represented by the following formula:

$$combied_features = \max(features_1, features_2) \tag{10}$$

In formulas (9) and (10), $features_1$ represents the image features, and $features_2$ represents the sensor features.

Under the condition that all other parameters and training processes remain unchanged, the same training set and testing set were used to evaluate the training results with different feature fusion methods.

This study collected a dataset comprising 100,000 video keyframes, with 20,000 keyframes representing each behavior. The dataset was divided into training and validation sets in a 9:1 ratio. During validation, each video keyframe sequence was set to a length of 16, corresponding to a time span of 4 s. Keyframes and sensor data sequences with fewer than 16 frames were excluded. The Accuracy, Precision, and Recall for the five behaviors using different feature fusion methods are detailed in Table 2. Specifically, Avg denotes mean fusion, Max denotes max fusion, and Cat denotes concatenation fusion. The model's optimal average performance on the validation set is as follows: an Accuracy of 98.80 %, a Precision of 97.15 %, and a Recall of 96.93 %. These results indicate that the proposed Transformer-based multimodal behavior recognition network can effectively identify the five basic movement behaviors of cows.

As illustrated in Table 2, the performance of the three fusion methods demonstrates that feature concatenation fusion outperforms max fusion, which in turn outperforms mean fusion. The mean fusion method averages each dimension of the two features, resulting in the smoothing of detailed information from both features. The max fusion method retains only the maximum values of the two features, causing the loss of relevant information from the other feature. Both fusion methods compress the original features, leading to the loss of some feature data and resulting in slightly lower performance compared to the feature concatenation fusion method. Although feature concatenation fusion preserves all feature information, allowing the model to utilize more information for decision-making during training, it also increases the computational complexity of the model.

As shown in Table 2, the performance of the cow behavior model based on multimodal data varies across different behaviors. The classification results for feeding behavior are the most accurate, with accuracy, precision, and recall all reaching 100 %, demonstrating the model's high accuracy and reliability in detecting feeding behavior. In contrast, while drinking behavior exhibits extremely high precision (100 %), the recall is only 87.93 %, indicating some missed detections. For lying and walking behaviors, the model can completely identify all actual behaviors (both recall rates are 100 %). However, the precision is slightly lower, at 93.85 % and 96.72 %, respectively, indicating a small number of false positives. The classification of standing behavior also shows high accuracy (98.33 %), precision (95.16 %), and recall (96.72 %), though slightly lower than other behaviors. Overall, the cow behavior classification model based on multimodal data performs excellently, with an average accuracy of nearly 99 %, and both precision and recall close to 97 %, demonstrating its efficiency and reliability in practical applications.

Table 2
Prediction Results of the Transformer Multimodal Network Based on Different Fusion Methods.

Behaviors	Number of sequence (s)	Funsion Type	Accuracy (%)	Precision (%)	Recall (%)
Drinking	58	Avg	97.67	100.00	87.93
		Max	97.67	100.00	87.93
		Cat	97.67	100.00	87.93
Feeding	61	Avg	99.67	98.39	100.00
		Max	100.00	100.00	100.00
		Cat	100.00	100.00	100.00
Lying	61	Avg	100.00	100.00	100.00
		Max	100.00	100.00	100.00
		Cat	98.67	93.85	100.00
Standing	61	Avg	97.33	92.06	95.08
		Max	96.33	84.72	100.00
		Cat	98.33	95.16	96.72
Walking	59	Avg	97.33	90.48	96.61
		Max	98.67	100.00	93.22
		Cat	99.33	96.72	100.00
Average	60	Avg	98.40	96.18	95.92
		Max	98.53	96.94	96.23
		Cat	98.80	97.15	96.93

3.2. Analysis of Single-Modality data results

3.2.1. Classification results based on videos

In this study, features extracted by Efficient V2 S were used to train a 2-layer Transformer encoder. The model parameters included an expected input feature number of 128, 8 multi-head attentions, a learning rate of 0.001, the Adam optimizer, a batch size of 32, and 20 training epochs. The highest accuracy was achieved during the 17th epoch, with average metrics across the five categories being 94.93 % for accuracy, 87.39 % for precision, and 87.23 % for recall. The confusion matrix obtained from the validation set is shown in Fig. 5.

The confusion matrix indicates that the recognition performance for walking and standing behaviors is relatively poor. This is due to the similar postures of walking and standing. Additionally, during the water-drinking period, cows often lift their heads to rest, causing drinking behavior to be misclassified as lying and walking. Walking behavior is often misclassified as other behaviors due to its similarity in posture with other actions. Hence, identifying cow behavior solely through video data can make it difficult to distinguish certain behaviors.

3.2.2. Classification results based on sensors

This study employed a BiLSTM with 7 hidden layers, each containing 128 nodes. To prevent overfitting, the dropout rate was set to 0.5, and the L2 regularization coefficient (weight decay) to 0.0002. The input data comprised acceleration and angular velocity in the xyz directions, with a sequence length of 16, a batch size of 32, and 200 training epochs. The highest accuracy appeared in the 17th epoch, with average metrics across the five categories being 96.83 % for accuracy, 92.02 % for precision, and 92.07 % for recall. The confusion matrix obtained from the validation set is shown in Fig. 6.

The confusion matrix shows that lying behavior is often mistakenly recognized as walking. This misclassification occurs because cows sometimes shake or lift their heads while lying, actions similar to their neck movements when walking. Lying is also misclassified as standing due to the still neck posture common in both behaviors. Additionally, there are instances where standing is misclassified as lying or walking and walking as drinking, feeding, lying, and standing, owing to the similarity in neck movements among these behaviors, making it challenging to distinguish them based solely on sensor data.

3.3. Behavior classification results based on multimodal data

This study combines video features with sensor features and trains a 2-layer Transformer encoder. The model parameters included an expected input feature number of 128, 8 multi-head attentions, a learning rate of 0.001, the Adam optimizer, a batch size of 32, and 20 training epochs. The highest accuracy appeared in the first epoch, and the confusion matrix obtained from the validation set is shown in Fig. 7.

The confusion matrix reveals that recognition performance for standing and walking is significantly better compared to using video data alone. The introduction of sensor data compensates for the deficiencies in recognizing visually similar behaviors, thereby improving the differentiation of the five behaviors, especially standing and walking. Due to the short duration and high concealment of drinking behavior, the recall rate was not significantly improved. However, the model's overall performance became more balanced. On the other hand, by integrating video data features, the classification results for lying, standing, and walking behaviors improved substantially, particularly the recall rate for walking behavior, which increased from 80.83 % to 100 %. This indicates that the behavior classification method based on multimodal data can compensate for the shortcomings of using different modal data individually, resulting in significantly superior classification outcomes compared to those based on single-modality data.

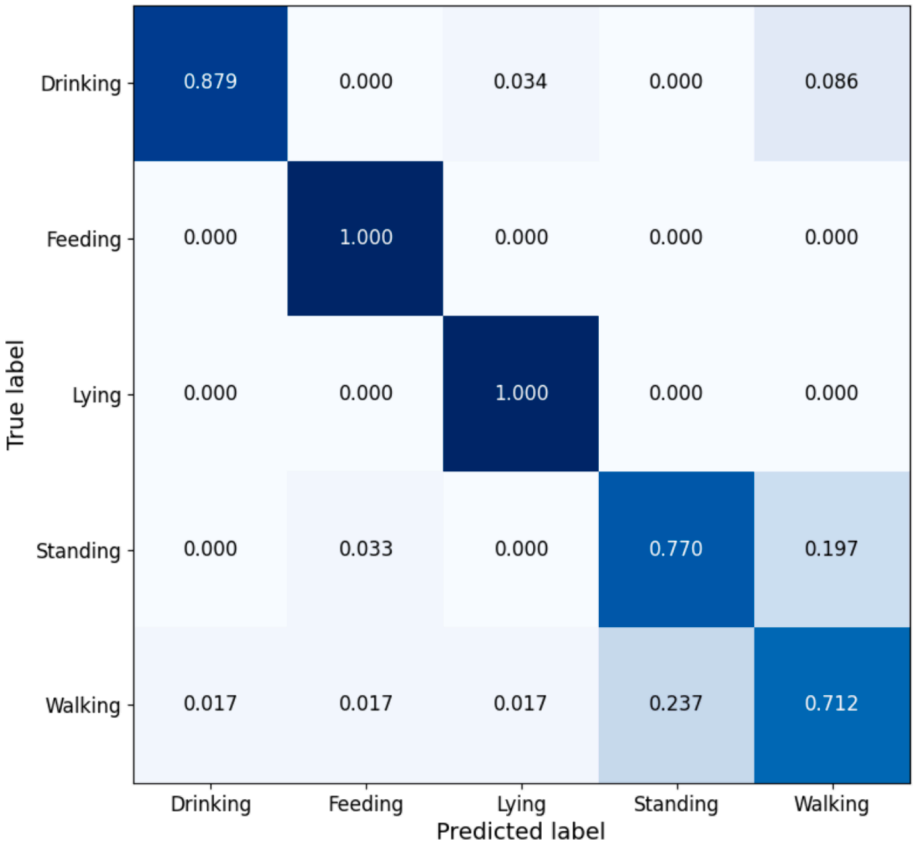


Fig.5. Confusion Matrix for Behavior Classification Based on Video Data.

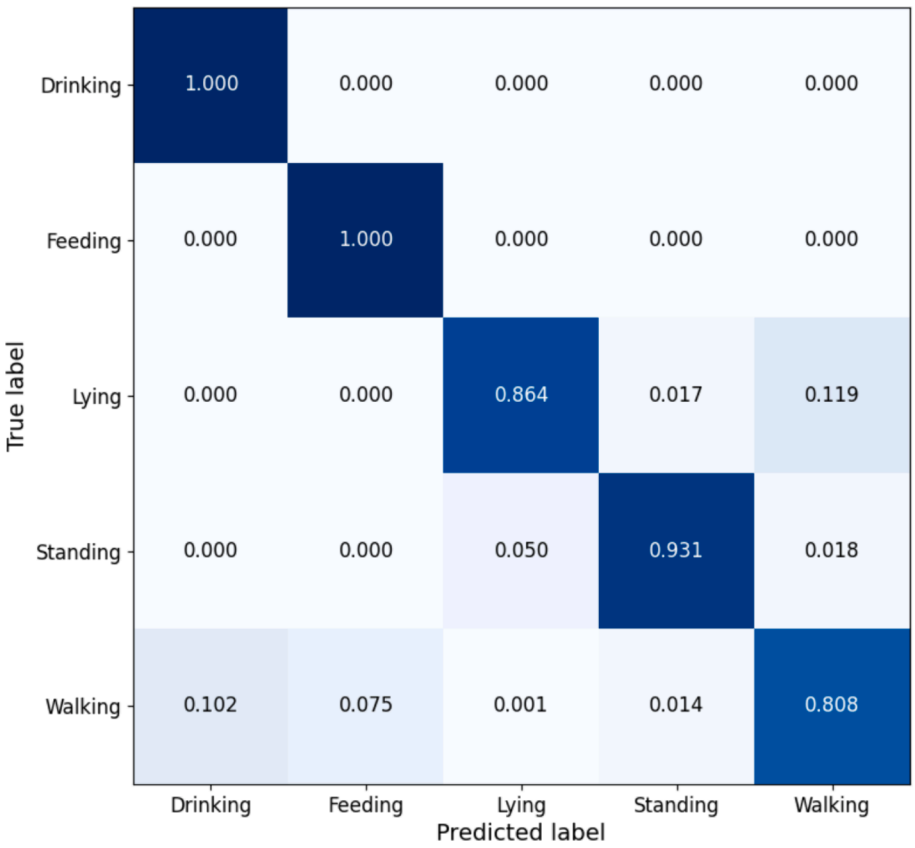


Fig. 6. Confusion Matrix for Behavior Classification Based on Sensor Data.

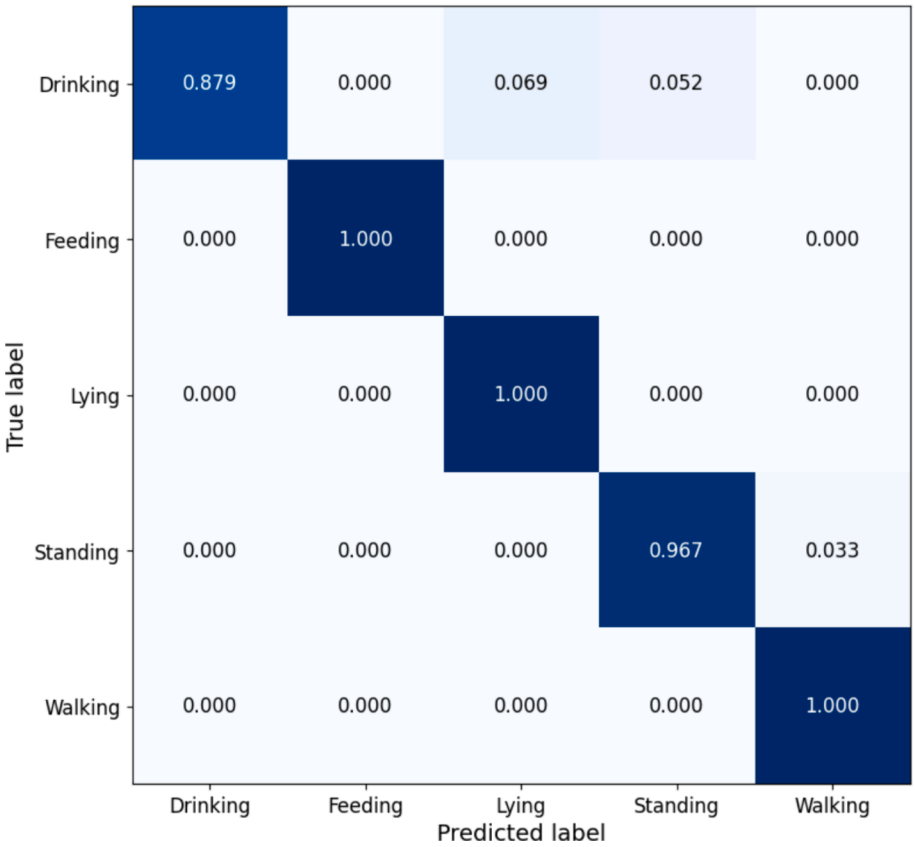


Fig. 7. Confusion Matrix for Behavior Classification Based on Multimodal Data.

3.4. Comparison of results with different sequence lengths

To determine the optimal sequence length for behavior recognition, this study conducted comparative experiments using five different sequence lengths: 8, 12, 16, 20, and 24. Consistent experimental conditions were maintained across all tests. Different sequence lengths were applied during feature extraction with BiLSTM for sensor data and during feature fusion for keyframe screenshot feature tensors and sensor feature vectors. The model that exhibited the best performance on the validation set was selected. Fig. 8 presents a line chart comparing the classification performance of models with different sequence lengths.

As shown in Fig. 8, accuracy increases with sequence length, peaking at a length of 16. Further increasing the sequence length results in a slight decline in accuracy, but it remains high. Precision and recall remain stable around 94 % for sequence lengths between 8 and 12. At a sequence length of 16, these metrics rise significantly to nearly 97 %. Further increases to sequence lengths of 20 and 24 result in a slight decrease but still perform above the initial level, approximately 96 %. When the sequence length is short (8 to 12), the model's performance is stable but relatively low, particularly in terms of recall. For longer sequence lengths (20 to 24), performance slightly declines but remains high, indicating good generalizability, albeit not optimal. Therefore, the model achieves the best performance at a sequence length of 16, attaining the highest accuracy, precision, and recall. This suggests that, at this length, the model effectively captures data features for accurate classification, balancing performance with computational resource efficiency.

3.5. Comparison with other methods

To assess the effectiveness of the models constructed using multimodal data for the classification of cattle behavior, this study utilized three different single-modality behavior classification methods. These methods were trained and validated on the dataset used in this research to evaluate the performance of the multimodal fusion method. The three methods are: classifying video data using 3D convolutional neural networks (3D CNN), performing feature extraction with EfficientNet V2 S followed by classification with a Transformer Encoder, and directly classifying sensor data using BiLSTM. The average classification accuracy, precision, and recall for each method are presented in Table 3:

As indicated in Table 3, the 3D CNN based on video data demonstrates high precision, suggesting that most predictions made as positive cases are accurate. However, its recall rate is lower, indicating the model is less effective at detecting all actual positive cases. The combination of

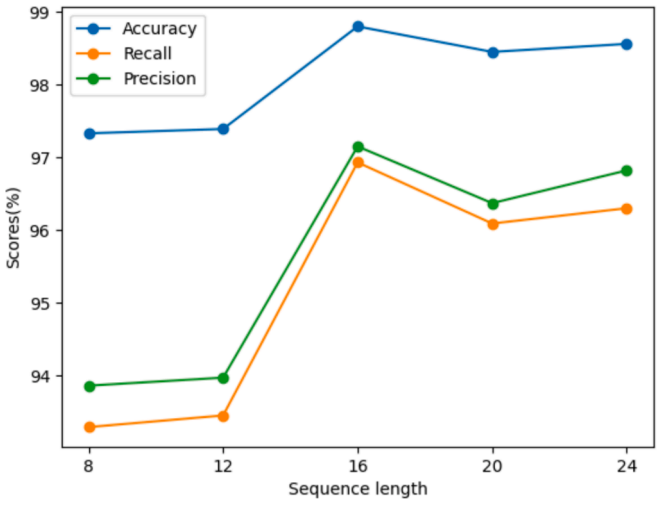


Fig. 8. Classification Performance Comparison of Models with Different Sequence Lengths.

Table 3
Comparison of Classification Results Using Different Methods.

Behaviors	Accuracy (%)	Precision (%)	Recall (%)
3D CNN	93.85	87.90	83.83
EfficientNet V2 S & Transformer	94.93	87.39	87.23
BiLSTM	96.83	92.02	92.07
Multimodal fusion method	98.80	97.15	96.93

EfficientNet V2 S and Transformer shows slightly higher accuracy and improved recall compared to the 3D CNN, indicating better identification of all actual positive cases. The BiLSTM model based on sensor data exhibits significant improvements across all metrics, particularly in accuracy and recall, suggesting it outperforms the previous two methods overall. The multimodal fusion method shows the best performance in all metrics, especially in precision and accuracy, indicating superior prediction of positive cases and overall classification correctness. Its high recall rate further demonstrates its effectiveness in detecting all actual positive cases. The performance comparison of the four classification methods is illustrated in Fig. 9:

In summary, while BiLSTM performs well across all metrics, it is slightly outperformed by the multimodal fusion method. Although EfficientNet V2 S & Transformer and 3D CNN also show commendable performance, they exhibit some deficiencies in accuracy and recall. Overall, the multimodal method that integrates video and sensor data demonstrates significant advantages in addressing complex behavior analysis tasks.

3.6. Discussion

3.6.1. Multimodal data analysis for cattle behavior

Shen et al. (Shen et al., 2020) employed tri-axial sensor data to identify cow behaviors, reporting a precision of 92.8 % for feeding and 93.7 % for rumination. While these results demonstrate the capability of single-modality sensor data, they also highlight its limitations in accurately capturing cow behaviors. In contrast, video data has shown a higher recognition accuracy of 97.87 % for basic cow behaviors. However, this method is more susceptible to disruptions caused by posture variations and environmental factors (Yin et al., 2020). By combining video and sensor data, a more holistic understanding of the video content can be achieved. Learning from different data modalities improves the performance of classification models. Multimodal approaches can address challenges that single-modal data cannot. For instance, non-contact cattle behavior perception is often hindered by occlusion, distant perspectives, and small targets. These limitations can be effectively mitigated by using sensor data as supplementary information. Conversely, the intrinsic behavior of cattle, such as momentary stillness, can make it difficult for sensor data to accurately distinguish behaviors. In such cases, the cattle's posture observed in videos can provide valuable insights into their state. Thus, the integration of video and sensor data through multimodal approaches enhances the comprehensiveness and depth of behavior information processing, thereby improving the model's understanding and classification capabilities. This is essential for solving complex behavior classification problems in cattle. Future research will consider further integration of other data modalities to develop a unified cattle detection model incorporating various data sources.

3.6.2. Model optimization issues

As an initial exploration into analyzing cattle behavior using multimodal data, this research has uncovered the potential for improved classification performance over single-modal data methods. However, the models used in this study still have room for optimization. In the stage of cow video feature extraction, how can the model be further optimized to efficiently and effectively extract key frames and behavioral features from the videos? Future research could focus on

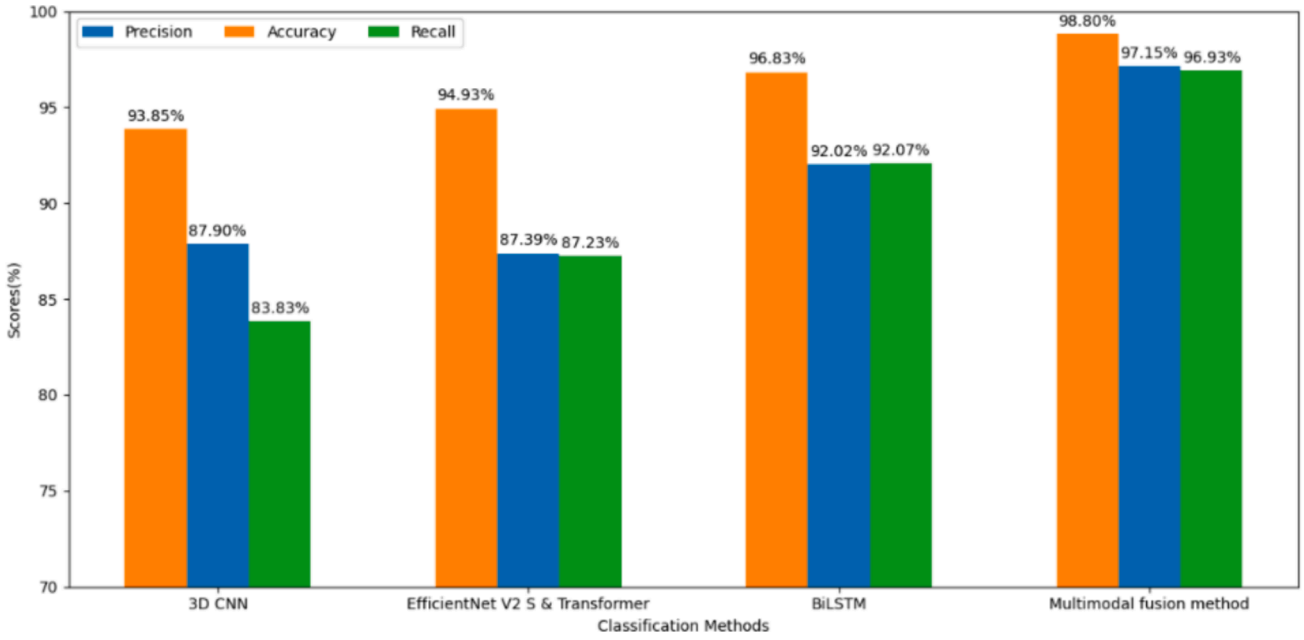


Fig. 9. Comparison of Classification Performance of Different Methods.

developing an end-to-end multimodal cow behavior recognition algorithm to simplify data processing workflows, enhance feature extraction efficiency, improve generalization capabilities, and strengthen model adaptability while reducing the risk of error propagation. Additionally, given the potential for partial modality data loss during data transmission in the cowshed environment, future studies could prioritize adaptive modality fusion techniques to address the challenge of temporary modality loss. Future research will progressively address these issues to make the model more applicable to practical production scenarios, improve the utilization of production equipment data, and achieve cost reduction and efficiency gains for farms.

4. Conclusion

This study implemented deep learning algorithms based on multimodal data to recognize cattle behaviors, including drinking, feeding, lying, standing, and walking. Compared to single-modality data behavior recognition methods, multimodal approaches offer several advantages, such as reduced stress, no limitations due to distant perspectives and occlusions, and the ability to leverage the complementary strengths of different data modalities. The results indicated that, with a sequence length of 16, the model achieved a precision of 97.67 % and a recall of 87.93 % for drinking behavior; a precision and recall of 100.00 % for feeding behavior; a precision of 98.67 % and a recall of 100.00 % for lying down behavior; a precision of 98.33 % and a recall of 96.72 % for standing behavior; and a precision of 99.33 % and a recall of 100.00 % for walking behavior. This method fully utilizes multimodal data collected from existing equipment to generate a detailed digital representation of cattle behaviors. This method enhances the model and simulation layer within the digital twin architecture for dairy cows, thereby laying a robust foundation for the development of comprehensive cattle behavior data models.

CRediT authorship contribution statement

Yi Zhang: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis. **Yu Zhang:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis. **Hailong Jiang:** Validation, Methodology, Data curation. **Haitao Du:** Validation, Resources. **Aidi Xue:** Validation. **Weizheng Shen:** Supervision, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was financially supported by the National Key R&D Program of China (Grant No. 2023YFD2000700); “Supported by the earmarked fund for CARS36”; The authors are grateful to anonymous reviewers for their comments.

References

Benaissa, S., Tuyttens, F.A.M., Plets, D., de Pessemier, T., Trogh, J., Tanghe, E., Martens, L., Vandaele, L., Van Nuffel, A., Joseph, W., Sonck, B., 2019. On the use of on-cow accelerometers for the classification of behaviours in dairy barns. Res. Veterinary Sci. 125, 425–433. https://doi.org/10.1016/j.rvsc.2017.10.005.

Chen, G.P., Li, C., Guo, Y., Shu, H., Cao, Z., Xu, B.B., 2022. Recognition of Cattle’s Feeding Behaviors Using Noseband Pressure Sensor With Machine Learning. Front. Vet. Sci. 9, 13. https://doi.org/10.3389/fvets.2022.822621.

Devlin, J., Chang, M.W., Lee, K., Toutanova, K., 2018. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding.

Fuentes, A., Yoon, S., Park, J., Park, D.S., 2020. Deep learning-based hierarchical cattle behavior recognition with spatio- temporal information. Comput. Electro. Agric. 177, 11. https://doi.org/10.1016/j.compag.2020.105627.

García, R., Aguilar, J., Toro, M., Pinto, A., Rodríguez, P., 2020. A systematic literature review on the use of machine learning in precision livestock farming. Comput. Electron. Agric. 179, 12. https://doi.org/10.1016/j.compag.2020.105826.

Halachmi, I., Guarino, M., Bewley, J., Pastell, M. 2019. Smart Animal Agriculture: Application of Real-Time Sensors to Improve Animal Well-Being and Production. in: *Annual Review of Animal Biosciences, Vol 7*, (Eds.) H.A. Lewin, R.M. Roberts, Vol. 7, Annual Reviews. Palo Alto, pp. 403-425.

Liu, N., Qi, J.W., An, X.P., Wang, Y., 2023b. A Review on Information Technologies Applicable to Precision Dairy Farming: Focus on Behavior, Health Monitoring, and the Precise Feeding of Dairy Cows. Agriculture-Basel. 13 (10), 21. https://doi.org/10.3390/agriculture13101858.

Liu, M., Wu, Y.Q., Li, G.Y., Liu, M.Q., Hu, R., Zou, H.W., Wang, Z.S., Peng, Y.Q., 2023a. Classification of cow behavior patterns using inertial measurement units and a fully convolutional network model. J. Dairy Sci. 106 (2), 1351–1359. https://doi.org/10.3168/jds.2022-22350.

Martin, M.J., Weigel, K.A., White, H.M., 2021. Assessment of the Relationship between Postpartum Health and Mid-Lactation Performance, Behavior, and Feed Efficiency in Holstein Dairy Cows. Animals 11 (5), 15. https://doi.org/10.3390/ani11051385.

Müller, R., Schrader, L., 2003. A new method to measure behavioural activity levels in dairy cows. Appl. Anim. Behav. Sci. 83 (4), 247–258. https://doi.org/10.1016/S0168-1591(03)00141-2.

Peter, A.T., Bosu, W.T., 1986. Postpartum ovarian activity in dairy cows: Correlation between behavioral estrus, pedometer measurements and ovulations. Theriogenology 26 (1), 111–115. https://doi.org/10.1016/0093-691x(86)90117-2.

Schuster, M., Paliwal, K.K., 1997. Bidirectional recurrent neural networks. IEEE Trans. Signal Process. 45 (11), 2673–2681.

Shen, W., Cheng, F., Zhang, Y., Wei, X., Fu, Q., Zhang, Y., 2020. Automatic recognition of ingestive-related behaviors of dairy cows based on triaxial acceleration. Inf. Process. Agric. 7 (3), 427–443. https://doi.org/10.1016/j.inpa.2019.10.004.

Stygar, A.H., Gomez, Y., Berteselli, G.V., Dalla Costa, E., Canali, E., Niemi, J.K., Llonch, P., Pastell, M., 2021. A Systematic Review on Commercially Available and Validated Sensor Technologies for Welfare Assessment of Dairy Cattle. Front. Veterinary Sci. 8, 15. https://doi.org/10.3389/fvets.2021.634338.

Tan, M., Le, Q.V., 2021. EfficientNetV2: Smaller Models and Faster Training.

Tian, F.Y., Wang, J., Xiong, B.H., Jiang, L.S., Song, Z.H., Li, F.D., 2021. Real-Time Behavioral Recognition in Dairy Cows Based on Geomagnetism and Acceleration Information. IEEE Access 9, 109497–109509. https://doi.org/10.1109/access.2021.3099212.

Tucker, C.B., Jensen, M.B., de Passillé, A.M., Hänninen, L., Rushen, J., 2021. Lying time and the welfare of dairy cows. J. Dairy Sci. 104 (1), 20–46. https://doi.org/10.3168/jds.2019-18074.

Wang, R., Gao, Z.Z., Li, Q.F., Zhao, C.J., Gao, R.H., Zhang, H.M., Li, S.Q., Feng, L., 2022. Detection Method of Cow Estrus Behavior in Natural Scenes Based on Improved YOLOv5. Agriculture-Basel. 12 (9), 19. https://doi.org/10.3390/agriculture12091339.

Wang, J., He, Z.T., Zheng, G.Q., Gao, S., Zhao, K.X., 2018. Development and validation of an ensemble classifier for real-time recognition of cow behavior patterns from accelerometer data and location data. PLoS One 13 (9), 19. https://doi.org/10.1371/journal.pone.0203546.

Yin, X.Q., Wu, D.H., Shang, Y.Y., Jiang, B., Song, H.B., 2020. Using an EfficientNet-LSTM for the recognition of single Cow's motion behaviours in a complicated environment. Comput. Electron. Agric. 177, 14. https://doi.org/10.1016/j.compag.2020.105707.

Zhang, Y., Zhang, Y., Gao, M., Dai, B.S., Kou, S.L., Wang, X.J., Fu, X., Shen, W.Z., 2023. Digital twin perception and modeling method for feeding behavior of dairy cows. Comput. Electron. Agric. 214, 14. https://doi.org/10.1016/j.compag.2023.108181.